

Boundary feedback control of a 2×2 weakly hyperbolic system: Lyapunov-based approach

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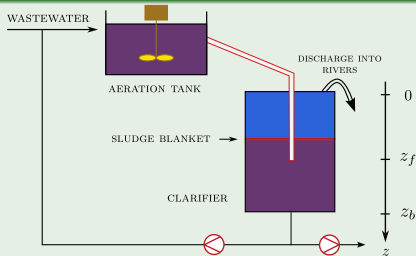
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Framework



1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: ε_s (particles density), $f_s := \varepsilon_s v_s$ (momentum) with v_s the particles velocity

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \frac{f_{1s}}{\rho_s} \delta_{z_f} \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = \underbrace{g \left(1 - \frac{\rho_l}{\rho_s} \right) \varepsilon_s}_{\text{Apparent weight}} + \underbrace{\frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)}}_{\text{Friction}} + \underbrace{\frac{f_{2s}}{\rho_s} \delta_{z_f}}_{\text{Feed}} \end{cases}$$

Two boundary conditions (**BCs**) at both ends of the spatial domain are needed to close the equations. These conditions are imposed by physical devices.

$$f_s(0, t) = U_0(t), \quad f_s(z_b, t) = U_1(t)$$

U_0, U_1 are *signal commands*.

—→ Coupling the equations with (**BCs**) corresponds to a *Boundary Control System*

Goal 1: Design a robust boundary controller that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth

Goal 2: Prove the stability of the controlled system

OUTLINE OF TALK

- 1 SYSTEM ANALYSIS
- 2 THROUGH A SIMPLIFIED CLARIFIER
- 3 NUMERICAL RESULTS
- 4 CONCLUSIONS AND PERSPECTIVES

SYSTEM ANALYSIS

SYSTEM ANALYSIS

Governing equations

$\mathcal{D} = [0, z_b] \times \mathbb{R}_+ /$ Find $\varepsilon_s, f_s: \mathcal{D} \rightarrow \mathbb{R}$ such that :

$$\begin{cases} \partial_t \varepsilon_s + \partial_z f_s = \frac{f_{1s}}{\rho_s} \delta_{z_f}, \\ \partial_t f_s + \partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right) = g \left(1 - \frac{\rho_l}{\rho_s} \right) \varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \frac{f_{2s}}{\rho_s} \delta_{z_f}, \\ f_s(0, t) = U_0(t), \quad f_s(z_b, t) = U_1(t), \\ \varepsilon_s(z, 0) = \varepsilon_s^0, \quad f_s(z, 0) = f_s^0. \end{cases} \quad (1)$$

with:

$$\sigma_e(\varepsilon_s) = \begin{cases} 0, & \varepsilon_s \leq \varepsilon_c \\ \sigma_0 \frac{\varepsilon_s^{n_s} - \varepsilon_c^{n_s}}{\varepsilon_c^{n_s}}, & \varepsilon_s > \varepsilon_c \end{cases}$$

ρ_l : water density ($:= 1000 \frac{\text{kg}}{\text{m}^3}$)

g : gravity acceleration constant ($:= 9.81 \frac{\text{m}}{\text{s}^2}$)

$\sigma_0, n_s, n_r, \varepsilon_c, \tau, \rho_s$ (parameters)

SYSTEM ANALYSIS

$$\underbrace{\partial_t \begin{pmatrix} \varepsilon_s \\ f_s \end{pmatrix}}_{\mathbf{Y}} + \underbrace{\partial_z \left(\frac{f_s^2}{\varepsilon_s} + \frac{\sigma_e(\varepsilon_s)}{\rho_s} \right)}_{\mathbf{F}(\mathbf{Y})} = \underbrace{\begin{pmatrix} 0 \\ g(1 - \frac{\rho_l}{\rho_s})\varepsilon_s + \frac{r(\varepsilon_s)(\varepsilon_s v_m - f_s)}{\rho_s \varepsilon_s (1 - \varepsilon_s)} + \frac{f_{2s}}{\rho_s} \delta_{zf} \end{pmatrix}}_{\mathbf{S}(\mathbf{Y})}$$

Quasi-linear form :

$$\partial_t \mathbf{Y} + \mathbf{A}(\mathbf{Y}) \cdot \partial_z \mathbf{Y} = \mathbf{S}(\mathbf{Y})$$

where

$$\mathbf{A}(\mathbf{Y}) := \frac{\partial \mathbf{F}}{\partial \mathbf{Y}} \in \mathcal{M}_2(\mathbb{R})$$

Eigenvalues of $\mathbf{A}(\mathbf{Y})$?

$$\text{Sp}(\mathbf{A}) = \left\{ v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s-1}}, v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s-1}} \right\}, \text{ if } \varepsilon_s > \varepsilon_c$$

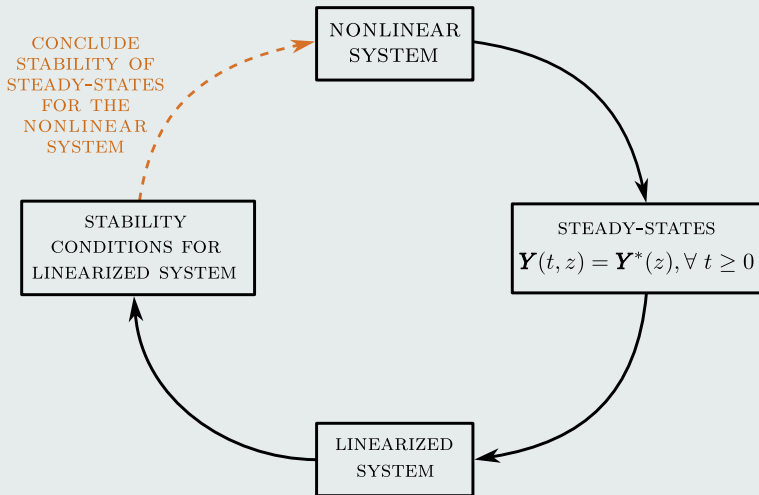
⇒ Strictly hyperbolic system

$$\text{Sp}(\mathbf{A}) = \{v_s, v_s\}, \text{ if } \varepsilon_s \leq \varepsilon_c$$

⇒ Weakly hyperbolic system

SYSTEM ANALYSIS

How to stabilize ? (Indirect Lyapunov's Method)



NONUNIFORM SHOCK STEADY-STATE

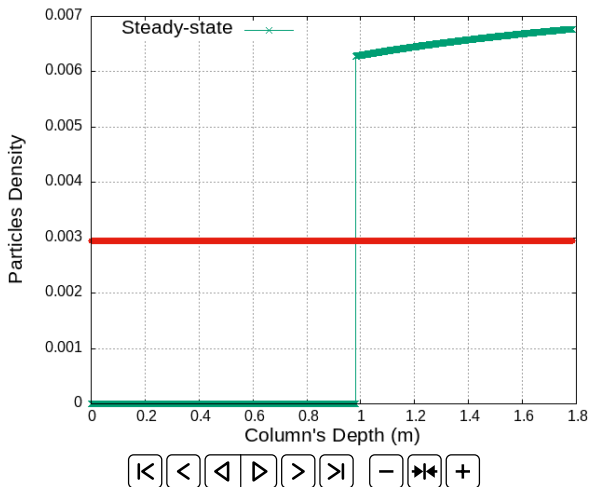


Fig. Simulation of the particles density on 6400 cells using Godunov scheme with an LLF Riemann Solver for flux computation: comparison with steady-state

SYSTEM ANALYSIS: LINEARIZATION

Consider small perturbations of the states with respect to steady-state:

$$E_s := \varepsilon_s(t, x) - \varepsilon_s^*(x), \quad V_s := v_s(t, x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \quad (2)$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma'_e(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

How to decouple system (2) ?

Determine a transformation that diagonalizes the matrix $\mathbf{A}^*(z)$?

Not uniformly possible (due to regions of weak-hyperbolicity)

Rather transform $\mathbf{A}^*(z)$ into a *Jordan* block:

$$\mathbf{A}^*(z) \longleftrightarrow \begin{pmatrix} c_1 & c_2 \\ 0 & c_3 \end{pmatrix}$$

THROUGH A SIMPLIFIED CLARIFIER

Target System

$$\begin{cases} \partial_t \phi(t, x) - A(x) \partial_x \phi(t, x) = B(x) \phi(t, x), & x \in (0, 1), t \geq 0 \\ \phi(0, x) = \phi^0(x) \end{cases} \quad (3)$$

where $\phi = (\phi_1, \phi_2)^\top : [0, +\infty) \times [0, 1] \rightarrow \mathbb{R}^2$ is the state.

The maps $A, B : [0, 1] \rightarrow \mathcal{M}_{2 \times 2}(\mathbb{R})$ are assumed $C^1([0, 1])$ and are defined as:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ B_{21}(x) & B_{22}(x) \end{pmatrix}$$

$c(x) \neq 0, \forall x \in [0, 1]$, and $\lambda_{1,2}(x) > 0$ s.t. $\exists x \in [0, 1]$ for which $\lambda_1(x) = \lambda_2(x)$.

Previous work

Assuming $B_{21}(x)=0$, and linear static boundary conditions specified as:

$$\phi(t, 1) = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix} \phi(t, 0) \quad (4)$$



M.O. AMIRAT, V. ANDRIEU, J. AURIOL, M. BAJODEK and C. VALENTIN, *Stability analysis of a class of 2 x 2 triangular hyperbolic systems*, IFAC-PapersOnLine

How to relax coefficients D_{21} and $B_{21}(x)$?

⇒ Design a wider class of controllers (of *Dynamical type*)

e.g.

$$\frac{d}{dt} \begin{bmatrix} \phi_1(t, 1) \\ \phi_2(t, 1) \end{bmatrix} = D \begin{bmatrix} \phi_1(t, 0) \\ \phi_2(t, 0) \end{bmatrix} + E \begin{bmatrix} \phi_1(t, 1) \\ \phi_2(t, 1) \end{bmatrix} + u(t)$$

where D and E are **full** matrices, and u is the *control*.

Boundary feedback control problem

$$\begin{cases} \partial_t \phi(t, x) - A(x) \partial_x \phi(t, x) = B(x) \phi(t, x), & x \in (0, 1), \\ \dot{\phi}(t, 1) = D \phi(t, 0) + E \phi(t, 1) + u(t), & t \in \mathbb{R}_+, \\ \phi(0, x) = \phi^0(x), & x \in (0, 1), \end{cases} \quad (5)$$

Sufficient conditions for the exponential stabilizability of system (5) when employing suitable control law, u ?

STABILITY ANALYSIS OF THE CLOSED-LOOP SYSTEM

State space

$X = L^2((0, 1); \mathbb{R}) \times H^1((0, 1); \mathbb{R}) \times \mathbb{R}^2$ equipped with the norm:

$$\begin{aligned} \|\cdot\|_X^2 : (f, g, z) \mapsto & \|f\|_{L^2((0,1),\mathbb{R})}^2 + \|g\|_{L^2((0,1),\mathbb{R})}^2 + \|g'\|_{L^2((0,1),\mathbb{R})}^2 \\ & + z^\top z \end{aligned}$$

Exponential stabilizability

The origin of system (5) is said to be globally δ -exponentially stabilizable in X , if there exists ν in $L_{loc}^\infty(\mathbb{R}_+)$, $\nu > 0$ and a decay rate $\delta > 0$ such that for any initial condition ϕ^0 satisfying compatibility conditions, we have:

$$\|\phi(t; \phi^0, \nu)\|_X^2 \leq \nu e^{-\delta t} \|\phi^0\|_X^2, \quad \forall t \in \mathbb{R}_+. \quad (6)$$

RESULTS

Recall:

$$A(x) = \begin{pmatrix} \lambda_1(x) & c(x) \\ 0 & \lambda_2(x) \end{pmatrix}, \quad B(x) = \begin{pmatrix} B_{11}(x) & B_{12}(x) \\ \textcolor{red}{B}_{21}(x) & B_{22}(x) \end{pmatrix}, \quad D \text{ and } E \text{ full matrices}$$

Assumption

There exists a function $\mathfrak{b} \in C^1([0, 1]; \mathbb{R})$ such that $\textcolor{red}{B}_{21}(x) = \mathfrak{b}(x)(\lambda_2(x) - \lambda_1(x))$.

Theorem

Let Assumption 1 and $\frac{\overline{\lambda'_2}}{\lambda_2} < \frac{1}{2}$ hold.

$\forall \kappa > 0$, there exist $\delta, \varepsilon_{\mathfrak{b}}, \varepsilon_B, \varepsilon_D > 0$ such that if,

$$\begin{aligned} \|D\|_2 &< \varepsilon_D, \\ \sup_{x \in (0,1)} \|B(x)\|_2 + \sup_{x \in (0,1)} \|B'(x)\|_2 &< \varepsilon_B, \\ \bar{\mathfrak{b}} + \bar{\mathfrak{b}}' &< \varepsilon_{\mathfrak{b}}, \end{aligned}$$

then the origin of the system (3) is δ -exponentially stabilizable in X with a control law given by

$$u(t) = -(\kappa \mathbb{I}_2 + E)\phi(t, 1). \quad (7)$$

SKETCH OF THE PROOF

The method used to prove the Theorem relies on the construction of a strict-Lyapunov functional (i.e. satisfying $\dot{\mathcal{V}}(\phi) \leq -\delta \mathcal{V}(\phi)$)

$$\mathcal{V} : \begin{cases} X & \rightarrow \mathbb{R}_+ \\ (\phi, z) & \mapsto (\mathcal{V}_1 + \mathcal{V}_2)(\phi) + \alpha(\mathcal{V}_3 + \mathcal{V}_4)(\phi) + \eta \mathcal{V}_5(z) \end{cases} \quad (8)$$

with $z = \phi(t, 1)$ and where

$$\begin{aligned} \mathcal{V}_1(\phi) &= \int_0^1 \frac{1 + \mu x}{\lambda_1(x)} \phi_1^2 dx, \quad \mathcal{V}_2(\phi) = \int_0^1 \frac{1 + \mu x}{\lambda_2(x)} \phi_2^2 dx \\ \mathcal{V}_3(\phi) &= \int_0^1 \frac{1 + \mu x}{\lambda_2(x)} (\partial_x \phi_2)^2 dx, \\ \mathcal{V}_4(\phi) &= 2 \int_0^1 \omega(x) \phi_1 \partial_x \phi_2 dx, \\ \mathcal{V}_5(z) &= z^\top z. \end{aligned}$$

parameters $\mu, \alpha, \eta \in \mathbb{R}_+^*$ and weight $\omega : [0, 1] \rightarrow \mathbb{R}$ will be selected later on.

SKETCH OF THE PROOF

Lemma 1 (Positivity, Coercivity and norm equivalence w.r.t state-space)

Let $e(\alpha)$ be given by

$$e(\alpha) = \frac{1}{2\alpha} \min \left\{ \frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \frac{\alpha}{\lambda_2} \right\} \quad (9)$$

If $\bar{\omega} \leq e(\alpha)$, there exist positive constants, $\nu_1(\alpha, \eta)$ and $\nu_2(\alpha, \eta)$, such that, for all (ϕ, z) in X

$$\nu_1(\alpha, \eta) \|(\phi, z)\|_X^2 \leq \mathcal{V}(\phi, z) \leq \nu_2(\alpha, \eta) \|(\phi, z)\|_X^2 \quad (10)$$

where

$$\nu_1(\alpha, \eta) = \min \left(\frac{1}{2\lambda_1}, \frac{1}{2\lambda_2}, \frac{\alpha}{2\lambda_2}, \eta \right), \quad \nu_2(\alpha, \eta) = \max \left(\frac{\frac{3}{2} + \mu}{\lambda_1}, \frac{1 + \mu}{\lambda_2}, \alpha \frac{\frac{3}{2} + \mu}{\lambda_2}, \eta \right)$$

SKETCH OF THE PROOF

Taking time derivative of each term of $\mathcal{V}$ along classical solutions of system (5), using integration by parts and performing Young Inequalities yields.

First,

$$\begin{aligned}\dot{\mathcal{V}}_2(\phi) &\leq (1 + \mu)\varepsilon_B \|\phi_1\|_{L^2}^2 \\ &\quad + \left(-\mu + 3(1 + \mu)\varepsilon_B \right) \|\phi_2\|_{L^2}^2 \\ &\quad + (1 + \mu)\phi_2^2(1) - \phi_2^2(0)\end{aligned}\tag{11}$$

Second,

$$\begin{aligned}\dot{\mathcal{V}}_1(\phi) &\leq \left(-\mu + (1 + \mu)(\bar{c}\gamma_1 + 3\varepsilon_B) \right) \|\phi_1\|_{L^2}^2 \\ &\quad + (1 + \mu)\varepsilon_B \|\phi_2\|_{L^2}^2 + (1 + \mu)\frac{\bar{c}}{\gamma_1} \|\partial_x \phi_2\|_{L^2}^2 \\ &\quad + (1 + \mu)\phi_1^2(1) - \phi_1^2(0)\end{aligned}\tag{12}$$

with $\gamma_1 > 0$.

SKETCH OF THE PROOF

Third,

$$\begin{aligned}
 \dot{\mathcal{V}}_3(\phi) &\leq (1 + \mu)(1 + \frac{\overline{\lambda'_2}}{\underline{\lambda_2}})\varepsilon_B(\|\phi_1\|_{L^2}^2 + \|\phi_2\|_{L^2}^2) \\
 &+ (1 + \mu)(\partial_x \phi_2(1))^2 - (\partial_x \phi_2(0))^2 \\
 &+ 2 \int_0^1 (1 + \mu x) B_{21}(x) \partial_x \phi_1(x) \partial_x \phi_2(x) dx \\
 &+ \left(-\mu + 2(1 + \mu) \left(\frac{\overline{\lambda'_2}}{\underline{\lambda_2}} + \varepsilon_B(2 + \frac{\overline{\lambda'_2}}{\underline{\lambda_2}}) \right) \right) \|\partial_x \phi_2\|_{L^2}^2
 \end{aligned} \tag{13}$$

Remark

The crossed term $\partial_x \phi_1 \partial_x \phi_2$ appearing in (13) cannot be handled using a Cauchy-Schwarz or Young inequality because the state ϕ_1 belongs to $L^2((0, 1); \mathbb{R})$ and not to $H^1((0, 1); \mathbb{R})$

SKETCH OF THE PROOF

Moreover, we have

$$\begin{aligned}
 \dot{\mathcal{V}}_4(\phi) &\leq \zeta(\varepsilon_B)(\bar{\omega} + \overline{\omega'}) \|\phi\|_X^2 \\
 &+ 2 \int_0^1 \omega(x)(\lambda_1(x) - \lambda_2(x)) \partial_x \phi_1(x) \partial_x \phi_2(x) dx \\
 &+ [B_{21}(x) \omega(x) \lambda_2(x) \phi_1^2(x)]_0^1 \\
 &+ 2[\omega(x) \lambda_2(x) \phi_1(x) \partial_x \phi_2(x)]_0^1 \\
 &+ [\omega(x) \lambda_1(x) B_{12}(x) \phi_2^2(x)]_0^1
 \end{aligned} \tag{14}$$

where $\zeta(\varepsilon_B) > 0$.

Last, we have,

$$\dot{\mathcal{V}}_5(z) = 2\eta \phi(t, 1)^\top \dot{\phi}(t, 1) = y^\top \begin{bmatrix} 0 & \eta D^\top \\ \eta D & -2\kappa\eta \end{bmatrix} y \tag{15}$$

where $y = [\phi(t, 0), \phi(t, 1)]^\top$.

SKETCH OF THE PROOF

Summing (11)-(12)-(13)-(14)-(15) we start by properly selecting the weight function $\omega(x)$ appearing in the functional $\mathcal{V}_4$ to ensure that the crossed term $\partial_x \phi_1 \partial_x \phi_2$ *vanishes*. Therefore, ω is determined such that,

$$2\alpha \int_0^1 [\omega(x)(\lambda_1(x) - \lambda_2(x)) + (1 + \mu x)B_{21}(x)] \partial_x \phi_1 \partial_x \phi_2 = 0$$

Using Assumption 1, we select

$$\omega(x) = (1 + \mu x)b(x). \quad (16)$$

SKETCH OF THE PROOF

We finally get:

$$\dot{\mathcal{V}}(\phi) \leq \underbrace{\mathcal{W}(\phi)}_{\text{Part without source term}} + \underbrace{(\varepsilon_B \tilde{\zeta} + (2 + 3\mu)\varepsilon_b \zeta(\varepsilon_B))\|\phi\|_X^2}_{\text{Part w source term}} + \underbrace{y^T M y}_{\text{Bdy terms}}, \quad (17)$$

where $\tilde{\zeta} > 0$,

$$\begin{aligned} \mathcal{W}(\phi) &= \left(-\mu + (1 + \mu)\bar{c}\gamma_1 \right) \|\phi_1\|_{L^2}^2 - \mu \|\phi_2\|_{L^2}^2 \\ &+ \left((1 + \mu)\frac{\bar{c}}{\gamma_1} + \alpha \left[-\mu + 2(1 + \mu)\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} \right] \right) \|\partial_x \phi_2\|_{L^2}^2, \end{aligned}$$

and,

$$M = \begin{bmatrix} K & G \\ \star & L \end{bmatrix} \in \mathcal{M}_{4 \times 4}(\mathbb{R})$$

SKETCH OF THE PROOF

Lemma 2

Let $\frac{\overline{\lambda_2'}}{\underline{\lambda_2}} < \frac{1}{2}$ holds and select

$$\mu > \max \left\{ 2, \frac{1 + 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}}{1 - 2\frac{\overline{\lambda_2'}}{\underline{\lambda_2}}} \right\} \text{ and } \alpha = (1 + \mu)^2 \bar{c}^2. \quad (18)$$

Then, there exists a positive scalar δ such that

$$\mathcal{W}(\phi) \leq -\delta(\|\phi_1\|_{L^2}^2 + \|\phi_2\|_{L^2}^2 + \|\partial_x \phi_2\|_{L^2}^2).$$

Then, we want to show that the quadratic form $y^\top M y$ is negative definite.

Approach (Schur Complement + Bound Estimates)

$$M \text{ negative definite} \Leftrightarrow \begin{cases} K \prec 0 \\ L - G^\top K^{-1} G \prec 0 \end{cases} \Leftrightarrow \begin{cases} \sigma_{\max}(K) < 0 \\ \sigma_{\max}(L - G^\top K^{-1} G) < 0 \end{cases}$$

SKETCH OF THE PROOF

Lemma 3

Let us define

$$m_1(\varepsilon_B) = \frac{1}{4\alpha s_1(\varepsilon_B)}, \quad m_2(\varepsilon_B, \bar{\mathbf{b}}) = \frac{\lambda_2(1)}{2\sqrt{\alpha(1+\mu)}}$$

$$m_3(\varepsilon_B, \bar{\mathbf{b}}) = \frac{1}{\sqrt{2} \left[\frac{\alpha(1+\mu)}{\lambda_2^2(1)} (\kappa + \lambda_2(1)\varepsilon_B) + \alpha \bar{\mathbf{b}} s_2(\varepsilon_B) + \eta \right]}$$

with $\eta = \frac{1}{2\kappa} \left(2 + \mu + \alpha \bar{\mathbf{b}} (s_2(\varepsilon_2) + s_3(\varepsilon_B)) + \alpha \frac{(1+\mu)}{\lambda_2^2(1)} (\kappa + \varepsilon_B \lambda_2(1))^2 \right)$.

If $\bar{\mathbf{b}} < m_1(\varepsilon_B)$ and $\varepsilon_D < \min(m_2(\varepsilon_B, \bar{\mathbf{b}}), m_3(\varepsilon_B, \bar{\mathbf{b}}))$ then $\exists \delta' > 0$ such that,

$$\mathbf{y}^\top M \mathbf{y} < -\delta' \|\phi(t, 1)\|_{\mathbb{R}^2}^2.$$

END OF THE PROOF

Based on these two Lemmas, the time derivative of the Lyapunov candidate functional along C^1 -solutions of system (5) satisfies

$$\dot{\mathcal{V}}(\phi) \leq -\tilde{\delta}\mathcal{V}(\phi). \quad (19)$$

where

$$\tilde{\delta} = \frac{\min(\delta, \delta') - (\varepsilon_B \tilde{\zeta} + (2 + 3\mu)\varepsilon_b \zeta(\varepsilon_B))}{\nu_2(\alpha, \eta)}$$

With Grönwall inequality the above equation (19) yields

$$\mathcal{V}(\phi(t; \phi_0)) \leq e^{-\tilde{\delta}t} \mathcal{V}(\phi^0), \quad \forall \phi^0 \in X.$$

To conclude, using the equivalence of norm stated in Lemma 1,

$$\|\phi(t; \phi^0)\|_X^2 \leq \frac{\nu_2}{\nu_1} e^{-\tilde{\delta}t} \|\phi^0\|_X^2$$

holds, which allows us to close the proof. ■

NUMERICAL RESULTS

EXAMPLE

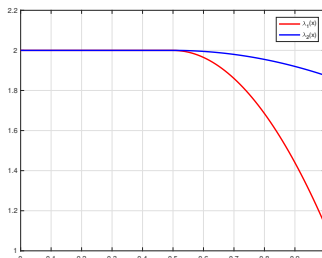
Consider system (5) with

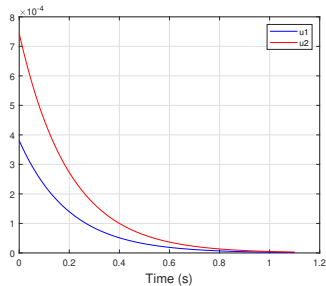
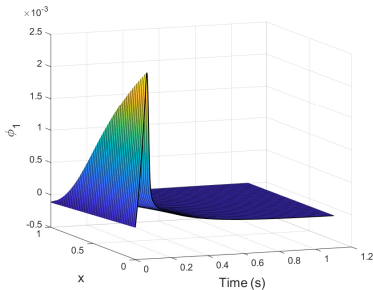
$$\lambda_1(x) = \begin{cases} 2 & \text{if } 0 \leq x < \frac{1}{2} \\ 2 - \frac{7}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$\lambda_2(x) = \begin{cases} 2, & \text{if } 0 \leq x < \frac{1}{2} \\ 2 + \frac{1}{2}(x - \frac{1}{2})^2 & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

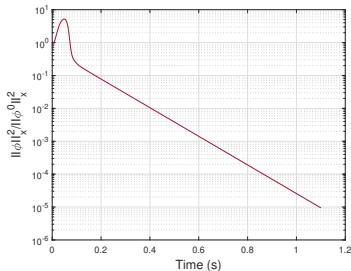
$$c(x) = x + 1, \quad D = \begin{bmatrix} \frac{1}{600} & \frac{1}{600} \\ \frac{1}{600} & \frac{1}{600} \end{bmatrix}, \quad \text{and } \kappa = 5. \quad B_{11}(x) = B_{12}(x) = B_{22}(x) = \frac{x}{3200},$$

$$B_{21}(x) = \frac{\lambda_2(x) - \lambda_1(x)}{3200}.$$





System (5): (left) space-time evolution of state ϕ_1 / (right) time evolution of boundary control $u(t)$



Time evolution of $\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$ in semi-logarithmic scale

CONCLUSIONS AND PERSPECTIVES

CONCLUSIONS

Study of a class of Non-Strictly Hyperbolic problems:

- Tackling the *new* difficulties related to the non-diagonal structure
- Establishing sufficient stability conditions for this class of systems
- Full Lyapunov analysis
- Numerical validation and comparison with theoretical results

Note that Non-Strictly Hyperbolic equations are of important interest and arise in many physical areas such as **Wastewater Treatment**, Elasticity theory, Advection of tracers in pipes etc.

Paper submissions

-  M.O. AMIRAT, V. ANDRIEU, J. AURIOL, M. BAJODEK and C. VALENTIN, *Stability analysis of a class of 2×2 triangular hyperbolic systems*, IFAC-PapersOnLine.
-  M.O. AMIRAT, V. ANDRIEU, J. AURIOL, M. BAJODEK and C. VALENTIN, *Boundary feedback control of a class of 2×2 weakly hyperbolic systems*, 5th IFAC Workshop on Control of Systems Governed by PDE (CPDE 2025).

PERSPECTIVES

Ongoing work

Drawback: Use of Cauchy-Schwarz/Young inequalities brings high conservatism in estimation of decay rates, overshoots in the Lyapunov analysis.

⇒ Relax conservatism by expressing stability conditions in terms of *feasability* of **Matrix Inequalities** (c.f. J.-M. CORON and G. BASTIN), *work in preparation...*

2) Apply this theory to the Clarifier/Aeration tank and conclude stability of the nonlinear system.

Thank you for your attention
Questions ?